

ADITYA KUMAR TIWARY

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PROFESSIONAL SUMMARY

AI-focused Software Engineering undergraduate with hands-on experience in Machine Learning, Deep Learning, NLP, Generative AI, and LLM-based applications. Skilled in building scalable backend systems, cloud-native microservices, and deploying end-to-end AI solutions using LangChain, RESTful APIs, and containerization.

TECHNICAL SKILLS

Languages & Databases: Python, C++, JavaScript, SQL, MongoDB, PostgreSQL, Redis

Frameworks & Libraries: Node.js, React.js, Next.js, Express.js, Django, FastAPI, TensorFlow, Keras, PyTorch, scikit-learn, Pandas, NumPy, Hugging Face Transformers, LangChain

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3, Lambda), CI/CD (GitHub Actions), Linux

Core CS & AI: Data Structures, Algorithms, OOP, Operating Systems, Computer Networking, Computer Vision, NLP, LLMs, RAG, Distributed Systems

PROFESSIONAL EXPERIENCE

AI & Cloud Intern — IBM

May 2024 – August 2024

- Developed and optimized a real-time traffic monitoring system using TensorFlow (computer vision), improving detection accuracy by 25%.
- Reduced API response latency by 30% through microservices optimization and inference pipeline tuning with Flask and Docker.
- Built a cloud-integrated dashboard handling 1000+ daily events for monitoring logs and system performance on AWS (EC2, S3).
- Deployed containerized services ensuring high availability and scalability with CI/CD pipelines.

PROJECTS

Smart News Analyzer (Distributed System)

Jan 2026 – Feb 2026

FastAPI, Django REST Framework, Next.js 14, React 18, TypeScript, PostgreSQL, Redis, Celery, NLP (BART, FinBERT, spaCy, Hugging Face), Scrapy, Sentry, JWT, Docker, AWS

- Engineered a full-stack distributed news analysis platform using Django REST Framework, Next.js 14, and FastAPI microservices, processing 100+ articles daily with real-time NLP-driven stock sentiment predictions.
- Implemented distributed task processing with Celery + Redis, supporting 1000+ concurrent API requests, achieving 99.9% uptime with automatic retry logic and error tracking.
- Developed transformer-based sentiment analysis pipeline (BART for summarization, FinBERT for financial sentiment) achieving 92% classification accuracy with GPU acceleration support.
- Built multi-source real-time data ingestion from 5+ platforms (Twitter/X, Telegram, YouTube, web scrapers) using Scrapy, sncrape, and async scrapers with rate limiting and deduplication.
- Optimized PostgreSQL database with full-text search indexing and Redis caching strategy, reducing API response time from 800ms to 150ms (5x improvement).
- Designed role-based access control (RBAC) with JWT authentication via NextAuth, supporting 1000+ users and personalized stock watchlists with notification system.
- Integrated Sentry error tracking and python-json-logger for production monitoring, reducing production issues by 40%.
- Created responsive admin dashboard with Chart.js for real-time market sentiment visualization and scraper configuration management.

AI Resume Analyzer (ATS Optimization Engine)

Sept 2025 – Oct 2025

Python, Django, React, NLP (BERT, TF-IDF)

- Created an AI-based system to parse and analyze resumes using BERT embeddings and keyword extraction.
- Developed a custom scoring algorithm improving keyword-job match accuracy by 40%.
- Designed backend APIs for handling concurrent requests and real-time processing with Django REST Framework.
- Implemented dynamic feedback system to optimize resumes based on job descriptions.

Lung Cancer Detection System (Deep Learning)

Jun 2025 – July 2025

TensorFlow, Flask, Docker, OpenCV

- Developed a CNN-based model with transfer learning for detecting lung cancer from CT scan images.
- Applied preprocessing and augmentation pipelines to improve model performance and generalization.

- Containerized model using Docker and deployed as a scalable REST API with Flask on AWS EC2.

EDUCATION

B.E. Electronics & Communication Engineering — [Sir M. Visvesvaraya Institute of Technology](#) 2022 – 2026

CERTIFICATIONS & ACHIEVEMENTS

- Solved 300+ DSA problems on LeetCode (Rating: 1616)
- 3rd Place – CertifyO Hackathon (AI-powered solution)
- Full Stack Development (MERN) – Udemy
- GATE C.S.E 2026 Qualified